# Adaptive Relationship Preserving Sparse NMF for Hyperspectral Unmixing

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***Abstract*— Hyperspectral unmixing is an essential research topic for spectral data analysis due to the existence of mixed pixels. Recently, many methods based on sparse nonnegative matrix factorization (NMF) have been widely used for unmixing by incorporating similarity preserving. However, most of them conduct the similarity learning and unmixing by using two separate steps, which may lead to the case that the learned similarity matrix is not the optimal one for unmixing. Thus, the performance of unmixing would be influenced and become undesirable. To alleviate this problem, we propose an adaptive relationship preserving-based sparse NMF (ARP-NMF) for hyperspectral unmixing. Typically, we regard similarity learning and unmixing as an alternative optimization process. During this process, the learned similarity weights and unmixing results can be mutually improved. Thus, our proposed method can learn the optimal similarity weights for unmixing and obtain better generalization ability for different hyperspectral images than traditional methods. Moreover, by using the spectral and spatial local structure, our ARP-NMF method effectively preserves structure consistency between pixels and abundances. Experimental results both on the synthetic data and the real data reveal that our proposed method outperforms several representative sparse NMF-based methods.**

***Index Terms*— Hyperspectral unmixing, nonnegative matrix factorization (NMF), similarity learning.**

## I. Introduction

HYPERSPECTRAL images provide several hundreds of contiguous spectral bands throughout the spectral region from the visible to the infrared, and they are important resources in the field of remote sensing [1], [2]. However, owing to the low spectral resolution of hyperspectral sensors and the high spatial complexity [3], most pixels in the captured hyperspectral images are "mixed pixels," which means that they are mixtures of several materials' reflectance values [4], [5]. The existence of mixed pixels brings challenges to many applications of the hyperspectral image, for example, food quality assessment [6], [7], environmental monitoring [8], and special area estimation [9]. Therefore, it is important to utilize hyperspectral unmixing as a preprocessing procedure for spectral data analysis. The purpose of unmixing is to estimate the spectral signatures included in mixed pixels and their corresponding proportion [10]–[12].

Among various hyperspectral unmixing methods, the linear mixture model (LMM) is one of the powerful and widely used tools [13]–[15]. LMM assumes that each mixed pixel can be approximatively expressed as the linear combination of pure spectral signatures, named endmembers, and weighted by their corresponding proportion, named fractional abundances [16], [17]. Generally, one mixed pixel does not include all endmembers in the hyperspectral scene but is a mixture of a few pure signatures [18]. That is to say, the most of entries in the abundance maps are zeros. Based on this cogitation, researchers have proposed a variety of sparse constrained unmixing algorithms [19], [20] and their variants [21]–[22].

In particular, the $L_0$ regularization is a straightforward constraint to define the sparsity. However, there are tremendous challenges because the $L_0$ minimization problem is a NP-hard problem. Therefore, many works utilize the constraint of $L_1$ or $L_{2,1}$ minimization to replace the $L_0$ minimization problem, which can be solved by the alternating direction algorithm for multipliers [24], [25]. For instance, Wang *et al.* [26] proposed a sparse redundant model by incorporating the redundant regularization into the traditional $L_1$ sparse regression formulation. Ma *et al.* [27] proposed a robust sparse unmixing method based on $L_{2,1}$ regularization to reduce the negative influence of noise and outliers. Moreover, many researchers utilize reweighted sparse regression to enhance the sparsity of the estimated fractional abundances [28], [29]. Under the $L_1$ framework, Zhang *et.al.* [30] proposed a spectral-spatial weighted sparse regression method to obtain a sparse solution constrained from both spectral and spatial domain. In addition, He *et al.* [31] incorporated a reweighted sparse regularization into nonnegative matrix factorization and assume that the numerical value of weights is inversely proportional to that of fractional abundances. Jia *et.al.* [32] adopted a reweighted $L_1$ norm minimization scheme to improve the estimation of abundances.

As we all know, hyperspectral unmixing includes endmember extraction and abundance evaluation. In the early time of unmixing research, these two parts are separately treated. For example, many geometry-based methods assume that all

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observations in one hyperspectral scene were located in the inside of a simplex, whose vertices exactly correspond to endmembers in this scene. These types of endmember extraction methods include vertex component analysis (VCA) [33], independent component analysis (ICA) [34] and so on [11], [35]. Furthermore, under two constraints of fractional abundance, i.e., abundance nonnegative constraint (ANC) and abundance sum-to-one constraint (ASC), the abundance maps are obtained [36].

Recently, NMF-based methods attract more attention for unmixing, which aim at decomposing a nonnegative matrix into the product of two low-rank nonnegative matrices [37], [38]. That is to say, NMF-based methods can deal with endmember extraction and abundance evaluation simultaneously. However, the traditional NMF algorithm has the drawback of the nonunique solution, because its cost function is non-convex. To deal with this problem, many works attempt to add reasonable constraints to obtain local optimal solutions.

As mentioned above, the sparse constraint is an appropriate option due to the character of spectral unmixing [39], [40]. Therefore, many researchers attempt to integrate sparsity constraint with NMF. For instance, Qian *et al.* [41] proposed a $L_{1/2}$-NMF algorithm, which verifies that $L_{1/2}$ regularization can bring sparser and more accurate unmixing results than $L_1$ regularization. And many extensions of $L_{1/2}$-NMF method are certified with advanced performance for unmixing. Inspired by the idea of deep learning, NMF is extended to the deep NMF model and used for unmixing. Fang *et al.* [42] unfold NMF into multilayers and add a sparsity constraint to each layer by using $L_1$ regularization. Feng *et al.* [43] took advantage of hierarchical features with hidden information and propose the sparsity-constrained deep NMF with total variation (SDNMF-TV) method. In addition, by describing the degree of diversity among substances, Yuan *et al.* [44] added the substance dependence constraint into $L_{1/2}$-NMF. In [45], a total variation regularization is imposed to utilize the structure of abundance maps, which has the benefit of reducing the influence of noise. Through adjusting the parameter of sparse regularization, Wang *et al.* [46] take advantage of the sparsity information and propose an effective algorithm based on adaptive half-thresholding. This method can obtain faster and sparser unmixing results than multiplicative update rule used in the standard NMF method.

In addition, spectral structure preserving is also integrated into sparse NMF. Many researchers prove that similar pixels always share similar abundances [47], [48]. Therefore, many works aim at maintaining the consistency of similarity between pixels and their abundances to improve unmixing performance [49], [50]. Specifically, Lu *et al.* [51] utilized the intrinsic manifold structure for spectral unmixing. Based on region segmentation, Liu *et al.* [52] designed local neighborhood weights based on spatial information to enhance spectral unmixing. However, these methods conduct the similarity learning and unmixing in two separate steps, which may bring about the consequence that the learned similarity matrix is not suitable for unmixing and these methods are susceptible to noise. Besides, clustering is used as a preprocessing tool for unmixing to learn the spectral structure of mixed pixels [53]–[55]. For instance, Wang *et al.* [54] integrated multiple clustering into the process of optimizing NMF to improve self-supervised unmixing performance. Khoshsokhan *et al.* [55] employed a clustered multitask network to model spectral unmixing problem, which has the advantage of node information in neighborhoods and clusters. These two methods use k-means and fuzzy c-means clustering and exploit different constraints on structure preserving. Note that clustering-based methods always divide pixels into a certain number of clusters. As we all know, most pixels in HSIs are mixed rather than pure. Therefore, clustering-based structure learning may lose the edge structure information, which is not conducive to improving the unmixing performance. Unlike these methods, in this article, we start from the viewpoint of finding a more suitable structure learning method to alleviate the aforementioned problems.

In this article, we propose an adaptive relationship preserving based sparse NMF method (ARP-NMF) for hyperspectral unmixing. Typically, we regard unmixing and similarity learning as a circulating optimization process. In this process, the similarity weights among pixels can be adaptively determined by unmixing results to some extent, and the updated similarity weights are used to guide the procedure of unmixing simultaneously. Thus, the learned similarity matrix always is the optimal one for unmixing, and our proposed method can obtain desirable generalization ability and robustness for diverse hyperspectral images. Moreover, the spectral and spatial local structures are integrated to improve the consistency of structure between pixels and abundances. Owing to that, the proposed method can obtain better unmixing performance.

The contributions of our work are summarized as follows.

1) We propose an adaptive relationship preserving based sparse NMF method, which has superiority in keeping the consistency of structure between pixels and abundances.
2) The ARP-NMF method builds an adaptive weights matrix to represent the similarity among pixels precisely, and impels similarity learning and unmixing to improve mutually.
3) In the proposed method, spectral and spatial local structures are incorporated together to preserve the similarity of pixels to the estimated abundances exhaustively.
4) We conduct a series of experiments both on synthetic data and real hyperspectral images to show the effectiveness of the proposed method. It can be found that ARP-NMF method has better generalization ability for diverse hyperspectral images than several representative sparse NMF based methods.

The rest of this article is designed as follows. Related works about our ARP-NMF method are introduced in Section II. Section III describes the proposed method in detail. In Section IV, experiments on both synthetic data and real hyperspectral data are presented. Finally, we make a conclusion in Section V.

## II. Related Works

In this section, we briefly describe the related works about NMF-based hyperspectral unmixing methods, which include

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linear mixing model and sparse nonnegative matrix factorization. In the following subsections, matrices and vectors are denoted by bold uppercases and lowercases, respectively. For convenience, we denote a hyperspectral image (HSI) with $N$ pixels as $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N] \in \mathbb{R}^{L\times N}$, where $L$ is the number of spectral bands.

### A. Linear Mixing Model

LMM is an essential framework to deal with hyperspectral unmixing problem, and it assumes that spectra of distinct materials are not interfered with each other [56], [57]. Therefore, a mixed pixel can be approximately expressed by a linear combination of several pure spectral signatures, named endmembers, and weighted by their proportion, named fractional abundances. The linear mixing model is defined as

$$\boldsymbol{x}_i = \boldsymbol{A}\boldsymbol{s}_i + \boldsymbol{e}_i \quad (i = 1, 2, \ldots, N) \tag{1}$$

where $\boldsymbol{x}_i$ denotes the $i$th mixed pixel in the hyperspectral image $\boldsymbol{X}$, $\boldsymbol{A} = [\boldsymbol{a}_1, \ldots, \boldsymbol{a}_M] \in R^{L\times M}$ is the endmember matrix, $\boldsymbol{s}_i \in R^{M\times 1}$ denotes the abundance vector of pixel $\boldsymbol{x}_i$, and $\boldsymbol{e}_i \in R^{L\times 1}$ is the additive noise. Note that the number of endmembers $M$ is much less than the number of spectral bands in the hyperspectral image, i.e. $M \ll L$.

For a hyperspectral image $\boldsymbol{X}$, LMM can be further represented as

$$\boldsymbol{X} = \boldsymbol{A}\boldsymbol{S} + \boldsymbol{E} \tag{2}$$

where $\boldsymbol{S} \in R^{M\times N}$ denotes the matrix of abundances, and $\boldsymbol{E}$ is the reconstruction error.

Generally, the abundances in LMM should satisfy the nonnegative constraint (ANC) and the sum-to-one constraint (ASC), which are formulated as

$$\begin{aligned} &\text{ANC}: \ \boldsymbol{s}_{ij} \geq 0, \quad (i = 1, \ldots, M, j = 1, \ldots, N) \\ &\text{ASC}: \ \sum_{i=1}^{M} \boldsymbol{s}_{ij} = 1, \quad (j = 1, \ldots, N). \end{aligned} \tag{3}$$

### B. Sparse Nonnegative Matrix Factorization

NMF is a widely used method in the field of data analysis, especially when nonnegative constraint is considered. The purpose of NMF is to decompose a given nonnegative matrix $\boldsymbol{X} \in R^{L\times N}$ into the product of two low-rank nonnegative matrices, and NMF can be written by

$$\boldsymbol{X} \approx \boldsymbol{A}\boldsymbol{S} \tag{4}$$

where $\boldsymbol{A} \in R^{L\times M}$, $\boldsymbol{S} \in R^{M\times N}$, and $M < min(L, N)$. To obtain the solutions of (5), NMF model can be regarded as a minimization problem of the Euclidean distance between $\boldsymbol{X}$ and $\boldsymbol{A}\boldsymbol{S}$ with nonnegative constraints on $\boldsymbol{A}$ and $\boldsymbol{S}$. The objective function of this optimization problem is as follows:

$$\min_{\boldsymbol{A}\geq 0, \boldsymbol{S}\geq 0} f(\boldsymbol{A}, \boldsymbol{S}) = \frac{1}{2}||\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}||_F^2 \tag{5}$$

where the operator '$\geq$' means element-wise equal-or-greater relation.

Among various optimization algorithms of (5), multiplicative iterative algorithm (MIA), proved by Lee and Seung [37], is widely used because its low complexity and stable convergence. The multiplicative update rules are as follows:

$$\boldsymbol{A} \leftarrow \boldsymbol{A}. * \left(\boldsymbol{X}\boldsymbol{S}^T\right) ./ \left(\boldsymbol{A}\boldsymbol{S}\boldsymbol{S}^T\right) \tag{6}$$

$$\boldsymbol{S} \leftarrow \boldsymbol{S}. * \left(\boldsymbol{A}^T\boldsymbol{X}\right) ./ \left(\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{S}\right) \tag{7}$$

where operator ".*" and "./" represent the element-wise multiplication and division, respectively, and $(\cdot)^T$ denotes the matrix transpose operator.

However, the solution space of NMF model is very large and many local optimal solutions exist because the objective function of (5) is not convex with respect to $\boldsymbol{A}$ and $\boldsymbol{S}$ together. In order to solve this problem, [41] integrated $L_{1/2}$ sparsity constraint into NMF model, and the $L_{1/2}$-NMF method can obtain sparser and more accurate solutions than other extensions of NMF method. The model of $L_{1/2}$-NMF can be represented by

$$\min_{\boldsymbol{A}\geq 0, \boldsymbol{S}\geq 0} f(\boldsymbol{A}, \boldsymbol{S}) = \frac{1}{2}\|\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}\|_F^2 + \mu\|\boldsymbol{S}\|_{1/2} \tag{8}$$

where $\mu \geq 0$ weights the contribution of sparse regularization term, which is defined as

$$||\boldsymbol{S}||_{1/2} = \sum_{m,n=1}^{M,N} (s_{mn})^{1/2} \tag{9}$$

where $\boldsymbol{S} = [s_{mn}]_{m=1,n=1}^{M,N}$. According to [41], the updating rules of $\boldsymbol{A}$ and $\boldsymbol{S}$ are as follows:

$$\boldsymbol{A} \leftarrow \boldsymbol{A}. * \left(\boldsymbol{X}\boldsymbol{S}^T\right) ./ \left(\boldsymbol{A}\boldsymbol{S}\boldsymbol{S}^T\right) \tag{10}$$

$$\boldsymbol{S} \leftarrow \boldsymbol{S}. * \left(\boldsymbol{A}^T\boldsymbol{X}\right) ./ \left(\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{S} + \frac{\mu}{2}\boldsymbol{S}^{-\frac{1}{2}}\right) \tag{11}$$

where $\boldsymbol{S}^{-1/2}$ is a matrix, in which each entry corresponds to the reciprocal of the square root of each element in matrix $\boldsymbol{S}$.

## III. Proposed Method

In this section, we first introduce the proposed the ARP-NMF method, and then present its optimization algorithm and implementation issues. The proposed method has superiority in modeling the similarity of pixels and preserving them consistent with the relationship of abundances, which contributes to improving the ability of unmixing.

### A. Adaptive Relationship Preserving Based Sparse NMF Method

According to the analysis of LMM, similar pixels should share similar abundances [47], [51]. Therefore, many NMF-based methods impose the constraint that makes the similarity of abundances consistent with the similarity of pixels to promote the performance of unmixing. However, most of them

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utilize the Heat Kernel function to model the similarity degree between two pixels $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$, which is denoted as

$$w_{ij} = e^{-\frac{\|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2}{\sigma}}. \tag{12}$$

Note that the above similarity degree is determined by their Euclidean distance only and may be inaccurate because of the existence of noise. Moreover, the similarity learning procedure and unmixing are conducted in two separate steps. Therefore, the similarity matrix constructed in this way may not be suitable for unmixing. Inspired by the clustering method based on adaptive neighbors [58], we plan to learn an adaptive similarity matrix $\boldsymbol{W} \in \mathcal{R}^{N\times N}$ to alleviate this problem in this article, which consists of the similarity of any pair of pixels.

Let $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N] \in \mathbb{R}^{L\times N}$ denote a hyperspectral data matrix. For the $i$th pixel $\boldsymbol{x}_i$, we define $w_{ij}$ to represent the similarity degree with neighbor $\boldsymbol{x}_j (j \neq i)$. The similarity degree can be determined by minimizing

$$\sum_{j=1}^{N}\left(\frac{\lambda_1}{2}\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2 w_{ij} + \gamma\, w_{ij}^2\right)$$
$$s.t.\ w_{ij} \geq 0, \quad w_{ii} = 0, \boldsymbol{w}_i^T \mathbf{1} = 1 \tag{13}$$

where $\lambda_1$ and $\gamma$ are positive tradeoff parameters. In order to ensure that the order of entries in $\boldsymbol{W}$ is fixed, we still set the range of $j$ from 1 to $N$ rather than 1 to $N-1$. In addition, we set $w_{ii} = 0$ in (13) to avoid the trivial solution where $w_{ii} = 1$. It can be found from (13) that the similarity degree $w_{ij}$ will be assigned a larger value if the distance between $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ is smaller. We can obtain the weights among all pixels by solving

$$\min_{\boldsymbol{W}} \sum_{i,j=1}^{N}\left(\frac{\lambda_1}{2}\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2 w_{ij} + \gamma\, w_{ij}^2\right)$$
$$\text{s.t. } \forall i, w_{ij} \geq 0, \quad w_{ii} = 0, \quad \boldsymbol{w}_i^T \mathbf{1} = 1. \tag{14}$$

Furthermore, in order to build the relationship among abundances in accordance with the similarity degree among pixels, we impose the spectral and spatial local structure constraints simultaneously. On the one hand, from the similarity learning procedure, we can see that the matrix $\boldsymbol{W}$ focuses on representing the spectral structure of a hyperspectral image. For each element in the matrix $\boldsymbol{W}$, a larger one means that the two corresponding pixels are more similar, and vice versa. Therefore, the learned adaptive weights are used to constrain the similarity degree of abundances, which can preserve the spectral structure of hyperspectral images to abundances. On the other hand, we utilize a segmentation method [59] to divide the image into several regions. The motivation of this preprocessing can be explained by using Fig. 1. As shown in Fig. 1, the first row displays the original Jasper Ridge image and illustration of image segmentation. We can see from Fig. 1(b), the HSI is divided into many regions with different shapes and sizes. Fig. 1(c)–(f) shows the curves of pixels in four regions. It can be seen that pixels in the same region share similar spectral characteristics. Therefore, it is reasonable to force abundances that correspond to pixels in the same region to have similar characteristics. To implement this, we utilize the abundance corresponding to the mean value of the pixels in each region. Specifically, assuming there are $Q$ regions, and for the $q$th region there are $n_q$ pixels, $\boldsymbol{x}_t^q$ $(q = 1, \ldots, Q,\ t = 1, \ldots, n_q)$ denotes the $t$th pixel in the $q$-region. We regard the average value of pixels in each region as a mixed pixel, which can be decomposed into an endmember matrix and an abundance vector. i.e. $\boldsymbol{u}_q = (1/n_q)\sum_{i=t}^{n_q} \boldsymbol{x}_t^q$ and $\boldsymbol{u}_q \approx \boldsymbol{A}\boldsymbol{v}_q$. Thus, matrix $\boldsymbol{V}$ composed of $Q$ column vectors $\boldsymbol{v}_q$ $(q = 1, \ldots, Q)$, i.e. $V = [\boldsymbol{v}_1, \ldots, \boldsymbol{v}_Q]$, as an auxiliary variable is introduced.

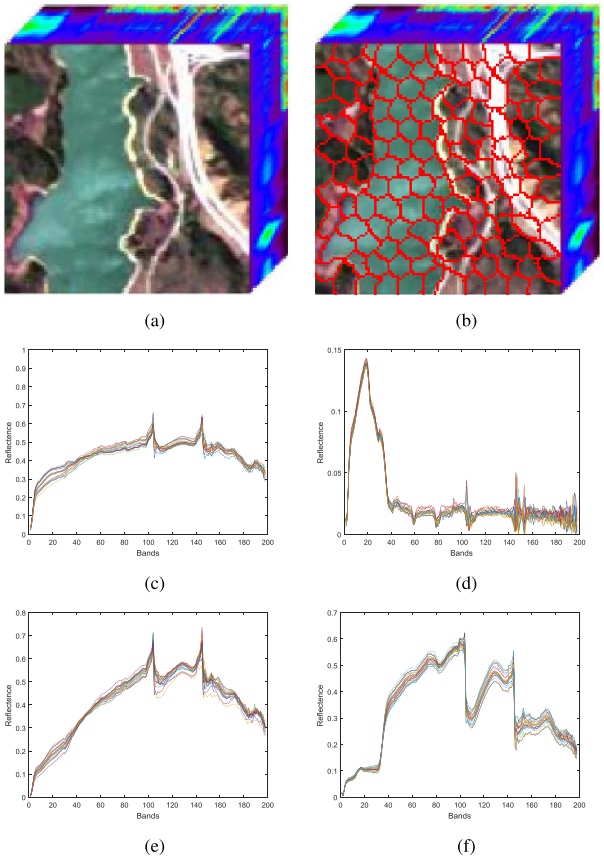



Fig. 1. Motivation of image segmentation. (a) and (b) Original Jasper Ridge image and the illustration of image segmentation, respectively. (c)–(f) Features of pixels in four regions.

Therefore, the spectral and spatial structure penalty term can be written as

$$\min_{S,W,V} \frac{\lambda_2}{2}\sum_{i,j=1}^{N}\|\boldsymbol{s}_i - \boldsymbol{s}_j\|_2^2 w_{ij} + \frac{\lambda_3}{2}\sum_{q,t=1}^{Q,n_q}\|\boldsymbol{s}_t^q - \boldsymbol{v}_q\|_2^2 \tag{15}$$

where $\lambda_2$ and $\lambda_3$ are positive tradeoff parameters, $\boldsymbol{s}_t^q$ denotes the abundance corresponding to the pixel $\boldsymbol{x}_t^q$, Obviously, the purpose of the first item in (15) is to make sure that the similarity structure of abundance is consistent with the spectral structure of pixels. The second item works for making abundances corresponding pixels in the same region share similar characteristics, and it can contribute to preserve the spatial local smoothness structure of pixels to abundances.

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Finally, the objective function of our ARP-NMF method can be expressed as

$$\begin{aligned}\min_{A,S,W,V} f(\boldsymbol{A},\boldsymbol{S},\boldsymbol{W},\boldsymbol{V}) &= \frac{1}{2}\|\boldsymbol{X}-\boldsymbol{A}\boldsymbol{S}\|_F^2+\mu\|\boldsymbol{S}\|_{1/2}\\ &+\sum_{i,j=1}^{N}\left(\frac{\lambda_1}{2}\|\boldsymbol{x}_i-\boldsymbol{x}_j\|_2^2 w_{ij}+\gamma\, w_{ij}^2\right)\\ &+\frac{\lambda_2}{2}\sum_{i,j=1}^{N}\|\boldsymbol{s}_i-\boldsymbol{s}_j\|_2^2 w_{ij}+\frac{\lambda_3}{2}\sum_{q,t=1}^{Q,n_q}\|\boldsymbol{s}_t^q-\boldsymbol{v}_q\|_2^2\\ s.t.\ \boldsymbol{A}\geq 0,\quad \boldsymbol{S}\geq 0,\ \boldsymbol{V}\geq 0,\ \boldsymbol{W}\geq 0,\ \boldsymbol{S}^T\mathbf{1}=\mathbf{1}\\ \boldsymbol{W}^T\mathbf{1}=\mathbf{1},\quad w_{ii}=0.\end{aligned} \tag{16}$$

### *B. Optimization Algorithm*

Obviously, the objective function (16) is not convex with respect to variables $\boldsymbol{A}$, $\boldsymbol{S}$, $\boldsymbol{W}$ and $\boldsymbol{V}$ together. Therefore, it is impracticable to acquire the global optimal solution. Alternatively, we employ a multiplicative iterative method to obtain local minimal solutions by optimizing the objective function alternately. The detailed procedures are described as follows.

*1) Optimization of A:* Due to the first two items in the objective function are related to $\boldsymbol{A}$, the updating rule of $\boldsymbol{A}$ is the same as $L_{1/2}$-NMF method, and it can be calculated by

$$\boldsymbol{A}\leftarrow\boldsymbol{A}.*\left(\boldsymbol{X}\boldsymbol{S}^T\right)./\left(\boldsymbol{A}\boldsymbol{S}\boldsymbol{S}^T\right). \tag{17}$$

*2) Optimization of V:* According to (16), only the last item is related to $\boldsymbol{V}$. Thus, the optimization problem of $\boldsymbol{V}$ in the vector view can be written as

$$\min_{\boldsymbol{v}_q\geq 0}\ \sum_{q,t=1}^{Q,n_q}\|\boldsymbol{s}_t^q-\boldsymbol{v}_q\|_2^2. \tag{18}$$

Obviously, it is a simple quadratic optimization problem and the optimal solution $\boldsymbol{v}_q$ can be calculated by

$$\boldsymbol{v}_q=\frac{1}{n_q}\sum_{t=1}^{n_q}\boldsymbol{s}_t^q. \tag{19}$$

From the updating rule of $\boldsymbol{v}_q$, we can see that each vector in matrix $\boldsymbol{V}$ is the mean of abundances corresponding to pixels in the related region. That is to say, the updating rule of $\boldsymbol{v}_q$ can ensure that $\boldsymbol{u}_q\approx\boldsymbol{A}\boldsymbol{v}_q$ is always true.

*3) Optimization of S:* Then, we consider the optimization problem of $\boldsymbol{S}$ in the vector form. Thus, it can be written as following problem with respect to $\boldsymbol{s}_t^q$

$$\begin{aligned}\min_{\boldsymbol{s}_t^q\geq 0} f\left(\boldsymbol{s}_t^q\right) &= \frac{1}{2}\|\boldsymbol{x}_t^q-\boldsymbol{A}\boldsymbol{s}_t^q\|_2^2+\mu\|\boldsymbol{s}_t^q\|_{1/2}\\ &+\frac{\lambda_2}{2}\sum_{j=1}^{N}\|\boldsymbol{s}_t^q-\boldsymbol{s}_j\|_2^2 w_{tj}+\frac{\lambda_3}{2}\|\boldsymbol{s}_t^q-\boldsymbol{v}_q\|_2^2.\end{aligned} \tag{20}$$

Its partial derivative with respect to $\boldsymbol{s}_t^q$ can be calculated by

$$\begin{aligned}\frac{\partial f}{\partial \boldsymbol{s}_t^q} &= -\boldsymbol{A}^T\boldsymbol{x}_t^q+\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{s}_t^q+\frac{\mu}{2}\left(\boldsymbol{s}_t^q\right)^{-\frac{1}{2}}+\lambda_2 d_{tt}\boldsymbol{s}_t^q\\ &-\lambda_2\boldsymbol{S}\boldsymbol{w}_t+\lambda_3\left(\boldsymbol{s}_t^q-\boldsymbol{v}_q\right)\end{aligned} \tag{21}$$

where $d_{tt}=\sum_j \boldsymbol{W}_{tj}$ is the $t$th diagonal element of diagonal matrix $\boldsymbol{D}$, and $\boldsymbol{w}_t$ is the $t$th column vector of symmetric matrix $\boldsymbol{W}$.

According to the gradient descent method, the abundance $\boldsymbol{s}_t^q$ can be updated by

$$\boldsymbol{s}_t^q\leftarrow\boldsymbol{s}_t^q-\boldsymbol{\eta}\frac{\partial f}{\partial \boldsymbol{s}_t^q} \tag{22}$$

where $\boldsymbol{\eta}$ is a step length vector. In order to eliminate the nonnegative items in the updating rule and maintain the nonnegativity of $\boldsymbol{s}_t^q$, $\boldsymbol{\eta}$ is defined as

$$\boldsymbol{\eta}=\boldsymbol{s}_t^q./\left(\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{s}_t^q+\frac{\mu}{2}\left(\boldsymbol{s}_t^q\right)^{-\frac{1}{2}}+\lambda_2 d_{tt}\boldsymbol{s}_t^q+\lambda_3\boldsymbol{s}_t^q\right). \tag{23}$$

Therefore, $\boldsymbol{s}_t^q$ can be updated by

$$\begin{aligned}\boldsymbol{s}_t^q\leftarrow\boldsymbol{s}_t^q.*&\left(\boldsymbol{A}^T\boldsymbol{x}_t^q+\lambda_2\boldsymbol{S}\boldsymbol{w}_t+\lambda_3\boldsymbol{v}_q\right)\\ &./\left(\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{s}_t^q+\frac{\mu}{2}\left(\boldsymbol{s}_t^q\right)^{-\frac{1}{2}}+\lambda_2 d_{tt}\boldsymbol{s}_t^q+\lambda_3\boldsymbol{s}_t^q\right).\end{aligned} \tag{24}$$

*4) Optimization of W:* Finally, when $\boldsymbol{A}$, $\boldsymbol{V}$ and $\boldsymbol{S}$ are fixed, the optimization problem of $\boldsymbol{W}$ can be written as

$$\begin{aligned}&\min_{\boldsymbol{W}}\sum_{i,j=1}^{N}\left[\left(\frac{\lambda_1}{2}\|\boldsymbol{s}_i-\boldsymbol{s}_j\|_2^2+\frac{\lambda_3}{2}\|\boldsymbol{x}_i-\boldsymbol{x}_j\|_2^2\right)w_{ij}+\gamma\, w_{ij}^2\right]\\ &\text{s.t. }\forall i,\quad \boldsymbol{w}_i^T\mathbf{1}=0,\ \boldsymbol{w}_i\geq 0.\end{aligned} \tag{25}$$

It can be seen from (25) that the optimization about the similarity matrix $\boldsymbol{W}$ is not only related to the distances between pixels but also determined by the distances between abundances. Therefore, the learned similarity matrix is more suitable for unmixing than traditional methods.

In addition, the problem of (25) is independent for each $i$, so we discuss how to optimize the following problem individually for each $\boldsymbol{w}_i$:

$$\begin{aligned}&\min\sum_{j=1}^{N}\left(\frac{\lambda_1}{2}\|\boldsymbol{s}_i-\boldsymbol{s}_j\|_2^2+\frac{\lambda_3}{2}\|\boldsymbol{x}_i-\boldsymbol{x}_j\|_2^2\right)w_{ij}+\sum_{j=1}^{N}\gamma_i w_{ij}^2\\ &s.t.\ \forall i,\quad \boldsymbol{w}_i^T\mathbf{1}=0,\ \boldsymbol{w}_i\geq 0\end{aligned} \tag{26}$$

where $\gamma_i$ is an auxiliary parameter to determine the value of $\gamma$, and detailed method is presented later. Let $\bar{d}_{ij}$ denote the linear combination of distances on pixels and abundances, it is defined as

$$\bar{d}_{ij}=\frac{\lambda_1}{2}\|\boldsymbol{s}_i-\boldsymbol{s}_j\|_2^2+\frac{\lambda_3}{2}\|\boldsymbol{x}_i-\boldsymbol{x}_j\|_2^2. \tag{27}$$

Then the Lagrangian function of problem (27) can be written by

$$\mathcal{L}\left(\boldsymbol{w}_i,\phi_i,\boldsymbol{\beta}_i\right)=\frac{1}{2}\|\boldsymbol{w}_i+\frac{\bar{\boldsymbol{d}}_i}{2\gamma_i}\|_2^2-\phi_i\left(\boldsymbol{w}_i^T\mathbf{1}-1\right)-\boldsymbol{\beta}_i^T\boldsymbol{w}_i \tag{28}$$

where $\phi_i$ and $\boldsymbol{\beta}_i$ are Lagrangian multipliers. The optimal solution $\boldsymbol{w}_i$ can be obtained by KKT condition as follows:

$$w_{ij}=\max\left(-\frac{\bar{d}_{ij}}{2\gamma_i}+\phi_i,0\right). \tag{29}$$

In order to alleviate the computation burden, it is reasonable to learn a sparse similarity matrix $\boldsymbol{W}$ [58]. Therefore, for each

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$\boldsymbol{w}_i$, only $k$ elements in it are non-zeros, which corresponding to the $k$ smallest values of $\bar{d}_{ij}(i \neq j)$. Without loss of generality, we utilize $w_{i1}, \ldots, w_{ik}$ to denote the $k$ non-zero elements arranged in descending order, and $\bar{d}_{i1}, \bar{d}_{i2}, \ldots, \bar{d}_{i(N-1)}$ are organized from small to large. According to (29) and the constraint $\boldsymbol{w}_i^T \mathbf{1} = 1$, we have

$$\phi_i = \frac{1}{k} + \frac{1}{2k\gamma_i} \sum_{j=1}^{k} \bar{d}_{ij} \tag{30}$$

where $\bar{d}_{ij}(j = 1, \ldots, k$, and $i \neq j)$ denote $k$ smallest values arranged in ascending order.

In addition, in order to make $\boldsymbol{w}_i$ contain exact $k$ non-zero entries, the constraint $w_{i,k+1} \leqslant 0 < w_{ik}$ must be satisfied. Then, combining it and (29), $\gamma_i$ can be set as

$$\gamma_i = \frac{k}{2}\bar{d}_{i,k+1} - \frac{1}{2}\sum_{j=1}^{k} \bar{d}_{ij}. \tag{31}$$

In addition, as set in [58], the overall $\gamma$ is set to the mean of $\gamma_i$. Then, by replacing $\gamma_i$ with $\gamma$ in (29), the solution of (25) can be obtained. Obviously, the value of $k$ is much easier to tune than the Lagrangian multipliers $\phi_i$ and the regularization parameter $\gamma$. In the later experiments, we set $k = 15$ in an empirical way.

In the end, the detailed optimization algorithm to solve the proposed ARP-NMF method is summarized in Algorithm 1.

**Algorithm 1** ARP-NMF Method for Hyperspectral Unmixing

**Input:**
The HSI matrix $\boldsymbol{X}$ and the number of endmembers $p$;
The regularization parameters: $\lambda_1$, $\lambda_2$, $\lambda_3$;
The stopping criteria parameters: $\epsilon$, $T$.
Others parameters: $\delta$, $k$, $Q$.
1: Initialize $\boldsymbol{A}$ and $\boldsymbol{S}$ by VCA and FCLS, respectively.
2: Segment the image and calculate the mean pixel $\boldsymbol{u}$ of each region.
**Repeat:**
3: Calculate the endmember matrix $\boldsymbol{A}$ by using Eq. (17).
4: Update abundance matrix $\boldsymbol{V}$ by solving problem (18).
5: Calculate the abundance vector $\boldsymbol{s}$ corresponding to each pixel by using Eq. (24).
6: Update similarity matrix $\boldsymbol{W}$ by solving problem (25).
7: **Until** either stop criterion is satisfied.
**Output:**
Endmember matrix $\boldsymbol{A}$;
Abundance matrix $\boldsymbol{S}$.

### C. Implementation Issues

In this section, we discuss some details in latter experimental part. The first one is how to ensure that two constraints in LMM are satisfied during the unmixing procedure. Obviously, from the updating rules of $\boldsymbol{S}$ and $\boldsymbol{V}$, we can conclude that ANC can be fulfilled as long as the initial $\boldsymbol{A}$ and $\boldsymbol{S}$ are nonnegative. And we replace $\boldsymbol{X}$ and $\boldsymbol{A}$ during each iteration by their augmented forms to guarantee ASC [36]. Their augmented forms are written by

$$\bar{\boldsymbol{X}} = \begin{bmatrix} \boldsymbol{X} \\ \delta \mathbf{1}_N^T \end{bmatrix}, \quad \bar{\boldsymbol{A}} = \begin{bmatrix} \boldsymbol{A} \\ \delta \mathbf{1}_M^T \end{bmatrix} \tag{32}$$

where $\mathbf{1}_N^T$ and $\mathbf{1}_M^T$ are row vectors with $N$-dimension and $M$-dimension, respectively, and $\delta$ is the tradeoff controling the effect of ASC in the objective function. According to [36], a larger $\delta$ results in a more precise ASC, but also leads to a slower convergence process. Therefore, we set $\delta = 15$ to balance these opposite effects.

Then, we use an unsupervised endmember extraction method, vertex component analysis (VCA) [33] to initialize $\boldsymbol{A}$, and employ fully constrained least squares (FCLS) [36] to obtain the initial $\boldsymbol{S}$.

Considering that some denominators may be zeros in the iteration process, we use a predefined threshold $10^{-2}$ to replace the entries in $\boldsymbol{V}$ and $\boldsymbol{S}$ when whose values are smaller than $10^{-2}$.

In addition, we provide two criteria to decide whether the procedure should be terminated. The first one is to predefine an error tolerance $\epsilon$, and in the later experimental part, we set $\epsilon = 0.01$. If the distance between two unmixing errors is smaller than $\epsilon$ for ten times successively, the iteration will be stopped. The stopping criterion can be represented by

$$Er\left(\boldsymbol{A}^t, \boldsymbol{S}^t\right) - Er\left(\boldsymbol{A}^{t-1}, \boldsymbol{S}^{t-1}\right) \leq \epsilon \tag{33}$$

where $Er(\boldsymbol{A}^t, \boldsymbol{S}^t) = (1/2)\|\boldsymbol{X} - \boldsymbol{A}^t\boldsymbol{S}^t\|_F^2$ and $Er(\boldsymbol{A}^{t-1}, \boldsymbol{S}^{t-1}) = (1/2)\|\boldsymbol{X} - \boldsymbol{A}^{t-1}\boldsymbol{S}^{t-1}\|_F^2$ denote the unmixing errors after $t$th and $(t-1)$th iteration, respectively. The other is to set a maximal number of iterations $T = 500$. In general, the updating procedure would be terminated when either stop criterion is satisfied.

The last essential point is how to choose parameters in our ARP-NMF method. Obviously, there are four regularization parameters $\mu$, $\lambda_1$, $\lambda_2$, and $\lambda_3$ in the objective function. We can see that the impact of parameter $\mu$ is to adjust the effect of sparsity regularization. It has been investigated in [60] that the parameter $\mu$ is related to the degree of sparsity, and it can be calculated by

$$\mu = \frac{1}{\sqrt{L}} \sum_{l=1}^{L} \frac{\sqrt{N} - \|\boldsymbol{x}^l\|_1/\|\boldsymbol{x}^l\|_2}{\sqrt{N-1}} \tag{34}$$

where $L$ and $N$ are the number of bands and pixels, respectively, and $\boldsymbol{x}^l$ denotes the $l$th row vector of matrix $\boldsymbol{X}$. In addition, parameters $\lambda_1$, $\lambda_2$, $\lambda_3$ will be discussed in the following experimental section. In addition, considering the motivation of image segmentation is to learn the local structure of HSIs, we set the number of regions $Q = \sqrt{N}$ in an empirical way, where $N$ is the number of pixels. In addition, from the summarization of Algorithm 1, we can see that the computational cost mainly in the iterative process. Then we analyze the floating-point calculation times of each step in iteration. The times of floating-point calculation for Steps 3, 4, and 5 are $P^2Q^2 + 2P^2(L + Q + 1) + 2PQ(L + 1)$, $P^2(2L-1) + 2PN(P + L))$ and $N(P^2 + P^2L + kP + PL + 4P)$. In addition, the cost of computing similarity matrix $\boldsymbol{W}$

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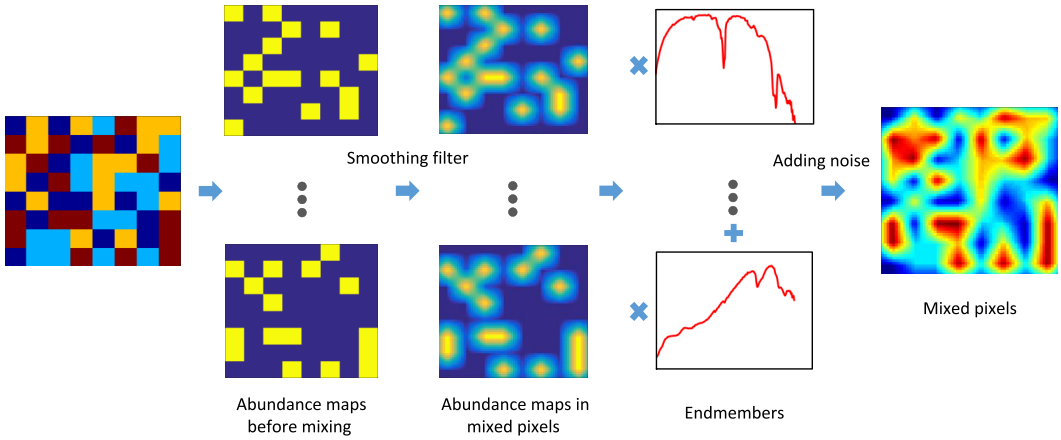



Fig. 2. Procedure of generating synthetic data.

TABLE I
COMPUTATIONAL COMPLEXITY OF THE SIX METHODS

| Method | Computational Complexity |
|---|---|
| $L_{1/2}$-NMF | $O(m(LPN + P^2N^2))$ |
| RCo-NMF | $O(m(NP^2))$ |
| GLr-NMF | $O(m(N^2P + NP^2 + LP^2 + LNP))$ |
| GL-NMF | $O(m(L + NP)NP + N^2L)$ |
| $HGL_{1/2}$-NMF | $O(mN^2(P^2 + 2P))$ |
| ARP-NMF | $O(m(N^2L + P^2(Q^2 + NL)))$ |

is $O(N^2L)$. Therefore, the overall cost of the ARP-NMF method is $O(m(N^2L + P^2(Q^2 + NL)))$ when the number of iterations is $m$. Table I lists the computational complexity of the ARP-NMF method and five representative algorithms.

## IV. EXPERIMENTS

In this section, we will investigate the performance of the proposed ARP-NMF method by comparing it with several representative methods, $L_{1/2}$-NMF [41], GL-NMF [51], $HGL_{1/2}$-NMF [47], RCo-NMF [61], and GLr-NMF [62]. These experiments are conducted on two types of hyperspectral data, the synthetic data and real hyperspectral data.

Moreover, we employ two criteria to measure the performance of unmixing, i.e. spectral-angle-distance (SAD) and root-mean-square-error (RMSE) [47], [51], [63]. They are defined as

$$\text{SAD} = \frac{1}{M}\sum_{i=1}^{M}\arccos\left(\frac{\boldsymbol{a}_i^T\widetilde{\boldsymbol{a}}_i}{\|\boldsymbol{a}_i\|\|\widetilde{\boldsymbol{a}}_i\|}\right) \tag{35}$$

$$\text{RMSE} = \left(\frac{1}{MN}\sum_{i=1}^{N}\|\boldsymbol{s}_i - \widetilde{\boldsymbol{s}}_i\|_2^2\right)^{\frac{1}{2}} \tag{36}$$

where $\boldsymbol{a}_i$ and $\widetilde{\boldsymbol{a}}_i$ mean the $i$th estimated endmember vector and corresponding ground-truth emdmember, respectively; $\boldsymbol{s}_i$ and $\widetilde{\boldsymbol{s}}_i$ denote the $i$th estimated abundance vector and the ground-truth abundance related to it, respectively.

It worth noticing that the ground-truth abundance maps for real hyperspectral data are not available. In this way, it is not feasible to assess the accuracy of estimated abundances by using (37). Therefore, we utilize $\text{RMSE}_X$ to quantify unmixing performance from the view of data reconstruction. It can be defined by

$$\begin{aligned}\text{RMSE}_X &= \frac{1}{\sqrt{MN}}\|\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}\|_F \\ &= \left(\frac{1}{MN}\sum_{i=1}^{N}\|\boldsymbol{x}_i - \widehat{\boldsymbol{x}}_i\|_2^2\right)^{\frac{1}{2}}\end{aligned} \tag{37}$$

where $\boldsymbol{X}$ denotes a hyperspectral data matrix, $\boldsymbol{x}_i$ and $\widehat{\boldsymbol{x}}_i$ denote the $i$th original pixel and re-constructed pixel, respectively. $\boldsymbol{A}$ and $\boldsymbol{S}$ denote the estimated endmember and abundance matrix, respectively. Note that we construct 20 tests for each experiment to get the average result. For simplicity, SAD and RMSE (or $\text{RMSE}_X$) in tables and figures denote the average SAD and average RMSE (or $\text{RMSE}_X$) in the following subsections.

### A. Synthetic Data

Before constructing experiments on the synthetic hyperspectral images, we introduce the procedure of generating synthetic data. We first randomly choose $P$ spectral signatures from a widely used spectral library, i.e., the USGS library, and regard the collection of $P$ signatures as the endmember set [47], [51]. Then, the process of generating synthetic data sized $d^2 \times d^2 (d \in \mathcal{Z}^+)$ consists of five steps: 1) dividing the image into $d \times d$ patches, and the size of each patch is $d \times d$; 2) padding each patch using the same endmember chosen randomly from endmember set. In this way, the abundance maps before mixing for all pure endmembers are generated; 3) generating mixed pixels by a $(d+1) \times (d+1)$ smoothing filter and make sure that the fractional abundances for pixels meet ASC constraint; 4) in order to guarantee endmembers have a reasonable contribution for each mixed pixel, we set

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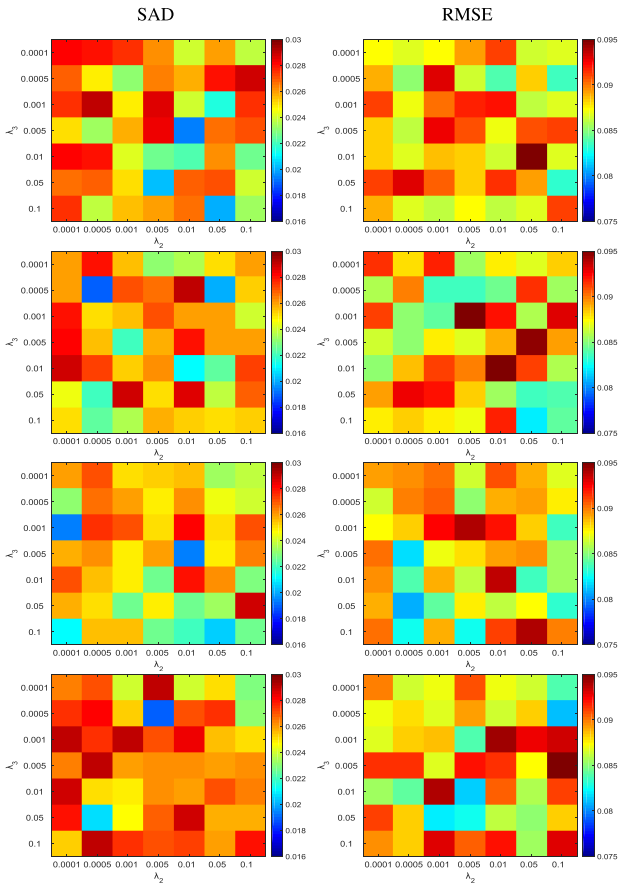



Fig. 3. Unmixing performance when parameters vary. In this figure, $\lambda_1$ equal to 0.0001, 0.001, 0.01, and 0.1 from the first row to the last in turn. The $x$-axis and $y$-axis represent $\lambda_2$ and $\lambda_3$, respectively, and colormaps in the first and second column represent the values of SAD and RMSE, respectively.

a parameter $\rho = 0.8$ to limit the purity of mixed pixels. In other words, if any element in one abundance map is larger than purity value $\rho$, all elements in this abundance will be replaced by $1/P$; and 5) the generated synthetic data is added zero-mean white Gaussian noise, whose signal-to-noise ratio (SNR) is set as 20 dB except the experiments about the robustness to noise. The procedure of generating synthetic data is shown in Fig. 2. It worth noticed that the illustration is not included in the fourth step.

*1) Sensitivity of Parameters:* In order to investigate the sensitivity of parameters, we construct a series of experiments when $\lambda_1$, $\lambda_2$, and $\lambda_3$ vary in the range of $\{1e^{-4}, 5e^{-4}, 1e^{-3}, 5e^{-3}, 1e^{-2}, 5e^{-2}, 0.1\}$. In these experiments, the synthetic data sized of $36 \times 36$ are generated by using four endmembers. And the other parameters of synthetic data are set as SNR $= 20$ dB and $\rho = 0.8$. For brevity, we show four groups of experiment results when $\lambda_1$ is different. Fig. 3 shows the SAD and RMSE when $\lambda_1$ is equal to $1e^{-4}$, $1e^{-3}$, $1e^{-2}$ and 0.1 from the first row to the last in turn. The $x$-axis and $y$-axis in each colormap represent $\lambda_2$ and $\lambda_3$, respectively. We can see that the values of SAD are almost in the range of 0.016 to 0.030 and the range of RMSE is approximately between 0.075 and 0.095. In other words, when three parameters in the proposed method are changed in the range of $\{0.0001, 0.0005, 0.001, 0.005, 0.01, 0.05, 0.1\}$, the fluctuations in unmixing performance are relatively small and irregular. Therefore, the proposed method is insensitive to changed parameters to some extent. Without loss of generality, we set $\lambda_1 = 0.01$, $\lambda_2 = 0.0005$, $\lambda_3 = 0.05$ in latter experiments to investigate other characteristics of the proposed ARP-NMF method.

*2) Robustness to Noise:* In this experiment, we aim at examining the robustness of six methods on corrupted synthetic images. The SNR of noise contained in each synthetic image is set from 15 to 55 dB with an interval of 10 dB. And all data used in this test include 4 endmembers. In addition, the other parameters of synthetic data are set as $\rho = 0.8$ and $d = 8$. Tables II and III demonstrate the average SAD and RMSE and standard deviation of six methods with different SNRs, respectively. In order to make the comparison clearer, we use a bold typeface to mark the best unmixing result. As shown, the performance of the six methods almost shares the same tendency that SAD and RMSE increase with the decreasing of SNR. Specifically, our ARP-NMF method has a better performance than the other five methods in most cases. For example, when SNR $= 15$ dB, the value of RMSE obtained by our ARP-NMF method is lower about 0.029 and 0.044 than those obtained by $L_{1/2}$-NMF and GLr-NMF method, respectively. With respect to the performance of endmember evaluation, the proposed ARP-NMF method is the best one except when SNR $= 35$ dB. The reason for this result may be that the low-rank constraint has the benefit of promoting the unmixing performance when SNR $= 35$ dB. But we can see that the distance between SAD of ARP-NMF and the lowest SAD, i.e. the SAD value of the GL-NMF method, is very small. Moreover, the standard deviation of SAD for ARP-NMF is much smaller than that for GL-NMF. That is to say, the stability of unmixing performance of ARP-NMF is better than GL-NMF.

*3) Number of Endmembers:* In this section, our purpose is to investigate the sensitivity of our ARP-NMF method when hyperspectral data consist of different numbers of endmembers. In addition, the synthetic images used in this experiment are added noise of SNR $= 20$ dB to make them more conform to the practical situation. And the other parameters are set as $\rho = 0.8$ and $d = 8$. Tables IV and V demonstrate the average SAD and RMSE and standard deviation when the number of endmembers in the range of 3–9, respectively. Obviously, the performance of almost all methods degenerates with the increasing number of endmembers. Meanwhile, the proposed method outperforms the other five methods in most cases except when the number of endmembers equals 7 and 9. This may be because the similarity learning of pixels will be slightly affected when the number of endmembers is larger. We can see that the difference between the unmixing performance of ARP-NMF and the best performance of compared methods in this case, i.e. that of the GL-NMF method, is very small. Moreover, the standard deviation of SAD and RMSE for ARP-NMF is smaller than other methods. That is to say, the proposed method has an obvious advantage in the stability of unmixing performance.

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TABLE II
SAD AND STANDARD DEVIATION OF THE SIX METHODS ON SYNTHETIC DATA WITH DIFFERENT SNRS

| SNR | $L_{1/2}$-NMF | RCo-NMF | GLr-NMF | GL-NMF | $HGL_{1/2}$-NMF | ARP-NMF |
|---|---|---|---|---|---|---|
| 55 | 0.0382±0.0104 | 0.0649±0.0132 | 0.0370±0.0096 | 0.0329±0.0116 | 0.0370±0.0137 | **0.0318±0.0043** |
| 45 | 0.0415±0.0113 | 0.1143±0.0103 | 0.0419±0.0113 | 0.0383±0.0122 | 0.0403±0.0114 | **0.0376±0.0054** |
| 35 | 0.0469±0.0106 | 0.1179±0.0129 | 0.0533±0.0132 | **0.0444±0.0134** | 0.0461±0.0121 | 0.0458±0.0044 |
| 25 | 0.0539±0.0195 | 0.1365±0.0193 | 0.0557±0.0108 | 0.0508±0.0150 | 0.0506±0.0105 | **0.0491±0.0036** |
| 15 | 0.0543±0.0217 | 0.1411±0.0186 | 0.0566±0.0196 | 0.0533±0.0219 | 0.0524±0.0148 | **0.0496±0.0069** |

TABLE III
RMSE AND STANDARD DEVIATION OF THE SIX METHODS ON SYNTHETIC DATA WITH DIFFERENT SNRS

| SNR | $L_{1/2}$-NMF | RCo-NMF | GLr-NMF | GL-NMF | $HGL_{1/2}$NMF | ARP-NMF |
|---|---|---|---|---|---|---|
| 55 | 0.0870±0.0203 | 0.1018±0.0222 | 0.0996±0.0144 | 0.0790±0.0157 | 0.0802±0.0186 | **0.0777±0.0077** |
| 45 | 0.0965±0.0172 | 0.1201±0.0132 | 0.0997±0.0175 | 0.0827±0.0199 | 0.0895±0.0172 | **0.0807±0.0066** |
| 35 | 0.1086±0.0239 | 0.1281±0.0169 | 0.1089±0.0253 | **0.0822±0.0222** | 0.1025±0.0194 | 0.0875±0.0051 |
| 25 | 0.1124±0.0217 | 0.1290±0.0172 | 0.1108±0.0179 | **0.0850±0.0169** | 0.1033±0.0204 | 0.0904±0.0067 |
| 15 | 0.1291±0.0244 | 0.1514±0.0214 | 0.1144±0.0212 | 0.1441±0.0286 | 0.1125±0.0226 | **0.1002±0.0082** |

TABLE IV
SAD AND STANDARD DEVIATION OF THE SIX METHODS ON SYNTHETIC DATA WITH DIFFERENT NUMBER OF ENDMEMBERS

| Number of Endmembers | $L_{1/2}$-NMF | RCo-NMF | GLr-NMF | GL-NMF | $HGL_{1/2}$NMF | ARP-NMF |
|---|---|---|---|---|---|---|
| 3 | 0.0404±0.0269 | 0.0545±0.0365 | 0.0346±0.0304 | 0.0415±0.0327 | 0.0426±0.0272 | **0.0310±0.0196** |
| 4 | 0.0414±0.0280 | 0.0643±0.0375 | 0.0423±0.0249 | 0.0458±0.0305 | 0.0439±0.0332 | **0.0351±0.0127** |
| 5 | 0.0421±0.0258 | 0.0778±0.0311 | 0.0454±0.0329 | 0.0484±0.0413 | 0.0446±0.0279 | **0.0403±0.0174** |
| 6 | 0.0471±0.0255 | 0.0780±0.0381 | 0.0514±0.0222 | 0.0506±0.0225 | 0.0459±0.0295 | **0.0422±0.0183** |
| 7 | 0.0500±0.0309 | 0.0889±0.0397 | 0.0549±0.0379 | 0.0527±0.0255 | **0.0478±0.0311** | 0.0490±0.0166 |
| 8 | 0.0544±0.0344 | 0.0933±0.0456 | 0.0656±0.0384 | 0.0625±0.0307 | 0.0519±0.0256 | **0.0503±0.0170** |
| 9 | 0.0576±0.0412 | 0.0990±0.0408 | 0.0663±0.0313 | 0.0657±0.0350 | **0.0548±0.0341** | 0.0571±0.0149 |

TABLE V
RMSE AND STANDARD DEVIATION OF THE SIX METHODS ON SYNTHETIC DATA WITH DIFFERENT NUMBER OF ENDMEMBERS

| Number of Endmembers | $L_{1/2}$-NMF | RCo-NMF | GLr-NMF | GL-NMF | $HGL_{1/2}$-NMF | ARP-NMF |
|---|---|---|---|---|---|---|
| 3 | 0.0974±0.0409 | 0.1126±0.0387 | 0.1009±0.0529 | 0.0978±0.0516 | 0.0969±0.0368 | **0.0911±0.0261** |
| 4 | 0.0995±0.0370 | 0.1236±0.0490 | 0.1074±0.0352 | 0.1014±0.0465 | 0.1074±0.0464 | **0.0921±0.0260** |
| 5 | 0.1021±0.0319 | 0.1303±0.0321 | 0.1176±0.0357 | 0.1058±0.0368 | 0.1090±0.0385 | **0.0987±0.0245** |
| 6 | 0.1071±0.0397 | 0.1323±0.0375 | 0.1131±0.0480 | 0.1071±0.0334 | 0.1117±0.0455 | **0.1006±0.0225** |
| 7 | 0.1118±0.0245 | 0.1341±0.0245 | 0.1218±0.0387 | 0.1149±0.0295 | 0.1148±0.0354 | **0.1020±0.0207** |
| 8 | 0.1130±0.0261 | 0.1358±0.0231 | 0.1300±0.0320 | 0.1160±0.0317 | 0.1156±0.0339 | **0.1076±0.0163** |
| 9 | **0.1139±0.0253** | 0.1375±0.0365 | 0.1316±0.0255 | 0.1181±0.0368 | 0.1158±0.0285 | 0.1151±0.0160 |

*4) Purity of Synthetic Data:* As mentioned above, we set parameter $\rho$ to make sure that each endmember has a reasonable contribution in the synthetic data. The purity of hyperspectral images controls the degree of mixing among endmembers, and the higher purity means the lower degree of mixing. In this experiment, our goal is to detect the influence of purity for estimating endmembers and abundances. We construct unmixing experiments when the purity value $\rho$ changes from 0.5 to 0.9 with an interval of 0.1. And the other parameters are set as SNR = 20 dB, $P = 4$ and $d = 8$. The average SAD and RMSE and standard deviation of six methods are shown in Tables VI and VII,

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TABLE VI
SAD AND STANDARD DEVIATION OF THE SIX METHODS ON SYNTHETIC DATA WITH DIFFERENT PURITY VALUES

| Purity values | $L_{1/2}$-NMF | RCo-NMF | GLr-NMF | GL-NMF | $HGL_{1/2}$-NMF | ARP-NMF |
|---|---|---|---|---|---|---|
| 0.5 | **0.1046±0.0353** | 0.1210±0.0373 | 0.1078±0.0349 | 0.1419±0.0398 | 0.1118±0.0340 | 0.1069±0.0209 |
| 0.6 | 0.0999±0.0325 | 0.1068±0.0473 | 0.1017±0.0329 | 0.1169±0.0524 | 0.0936±0.0291 | **0.0801±0.0226** |
| 0.7 | 0.0706±0.0302 | 0.0907±0.0483 | 0.0739±0.0261 | 0.0691±0.0223 | 0.0669±0.0293 | **0.0565±0.0154** |
| 0.8 | 0.0583±0.0239 | 0.0679±0.0323 | 0.0434±0.0230 | 0.0519±0.0207 | 0.0406±0.0309 | **0.0378±0.0165** |
| 0.9 | 0.0317±0.0159 | 0.0558±0.0171 | 0.0304±0.0125 | 0.2825±0.0141 | 0.0303±0.0167 | **0.0264±0.0076** |

TABLE VII
RMSE AND STANDARD DEVIATION OF THE SIX METHODS ON SYNTHETIC DATA WITH DIFFERENT PURITY VALUES

| Purity values | $L_{1/2}$-NMF | RCo-NMF | GLr-NMF | GL-NMF | $HGL_{1/2}$-NMF | ARP-NMF |
|---|---|---|---|---|---|---|
| 0.5 | 0.1808±0.0460 | **0.1602±0.0445** | 0.1731±0.0338 | 0.2049±0.0358 | 0.1847±0.0381 | 0.1648±0.0278 |
| 0.6 | 0.1682±0.0314 | 0.1594±0.0537 | 0.1617±0.0229 | 0.1980±0.0373 | 0.1792±0.0310 | **0.1537±0.0194** |
| 0.7 | 0.1431±0.0400 | 0.1355±0.0587 | 0.1449±0.0299 | 0.1535±0.0353 | 0.1482±0.0379 | **0.1260±0.0177** |
| 0.8 | 0.0961±0.0312 | 0.1213±0.0400 | 0.1155±0.0345 | 0.1144±0.0272 | 0.0906±0.0357 | **0.0884±0.0186** |
| 0.9 | 0.0630±0.0349 | 0.1117±0.0334 | 0.0985±0.0312 | 0.0818±0.0294 | 0.0625±0.0317 | **0.0602±0.0185** |

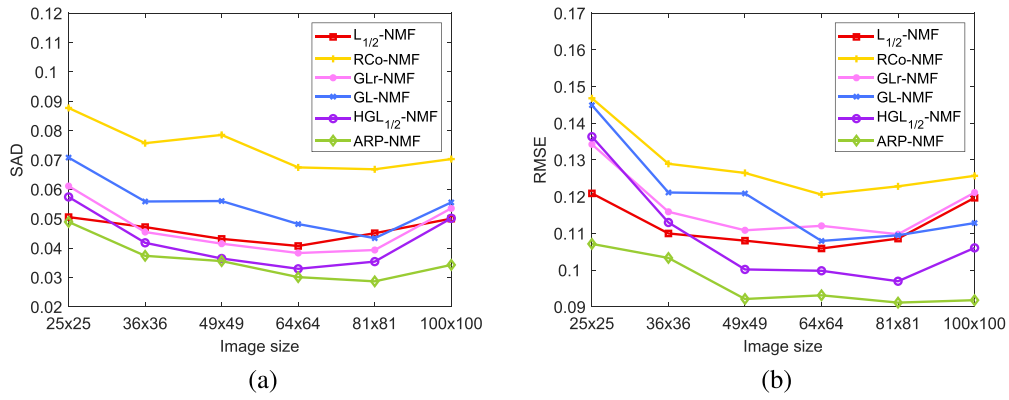



Fig. 4. (a) SAD and (b) RMSE of six algorithms on synthetic data with different image sizes.

respectively. It can be seen from Tables VI an d VII that SAD and RMSE of all methods decrease when the purity values of synthetic data increase and our ARP-NMF method obtains the best performance of abundance evaluation except the case of $\rho = 0.5$. The reason for this result may be that the proposed algorithm has a better advantage when the mixing degree is high, but the advantage is not obvious when the mixing degree is low.

*5) Image Size of Synthetic Data:* The purpose of this experiment is to investigate the sensitivity of the proposed method to different image sizes. The sizes of synthetic data used in this experiment are $25 \times 25$, $36 \times 36$, $49 \times 49$, $64 \times 64$, $81 \times 81$ and $100 \times 100$, respectively. The number of endmembers in synthetic data is 4, and the other parameters are set as SNR = 20 dB and $\rho = 0.8$. Fig. 4 demonstrates the unmixing performance of six algorithms. As shown in Fig. 4, the proposed ARP-NMF method achieves the best unmixing performance on all synthetic images of different sizes. In addition, we can see that the unmixing performance of most algorithms can be divided into two stages as the number of pixels increases. At first, when the image size increases from $25 \times 25$ to $64 \times 64$, the unmixing performance of the six algorithms is gradually improved. This is because it is difficult for NMF-based algorithms to obtain sufficient statistical information when the number of samples is too small. Then as the image size increases, the unmixing performance of some methods will deteriorate slightly, for example, $HGL_{1/2}$-NMF, $L_{1/2}$-NMF and GLr-NMF method. This may be because the enlargement of image size will reduce the possibility of finding the optimal solution. Moreover, the performance of the ARP-NMF method gradually stabilizes rather than deteriorates as the number of pixels increases.

### *B. Real Hyperspectral Data*

In this subsection, we perform experiments on three popular real hyperspectral data sets to investigate the unmixing performance, i.e. Samson, Jasper Ridge, and AVIRIS Cuprite. They are shown in Fig. 5.

*1) Samson Data:* Samson is the first public real hyperspectral data set on a website. This image contains

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TABLE VIII
SAD AND STANDARD DEVIATION OF THE SIX METHODS ON SAMSON DATA

| Endmember | $L_{1/2}$-NMF | RCo-NMF | GLr-NMF | GL-NMF | $HGL_{1/2}$-NMF | ARP-NMF |
|---|---|---|---|---|---|---|
| Tree | 0.0282±0.0142 | 0.0498±0.0027 | 0.0944±0.0304 | 0.0254±0.0148 | 0.0259±0.0175 | **0.0189±0.0061** |
| Water | 0.0572±0.0163 | 0.0913±0.0069 | 0.0770±0.0098 | 0.0463±0.0136 | 0.0516±0.0145 | **0.0436±0.0051** |
| Soil | 0.2314±0.0264 | 0.3708±0.0075 | 0.2511±0.1202 | 0.1781±0.0320 | 0.2391±0.0252 | **0.1103±0.0151** |
| Mean | 0.1056±0.1011 | 0.1706±0.1749 | 0.1408±0.1060 | 0.0833±0.0654 | 0.1055±0.0960 | **0.0576±0.0343** |

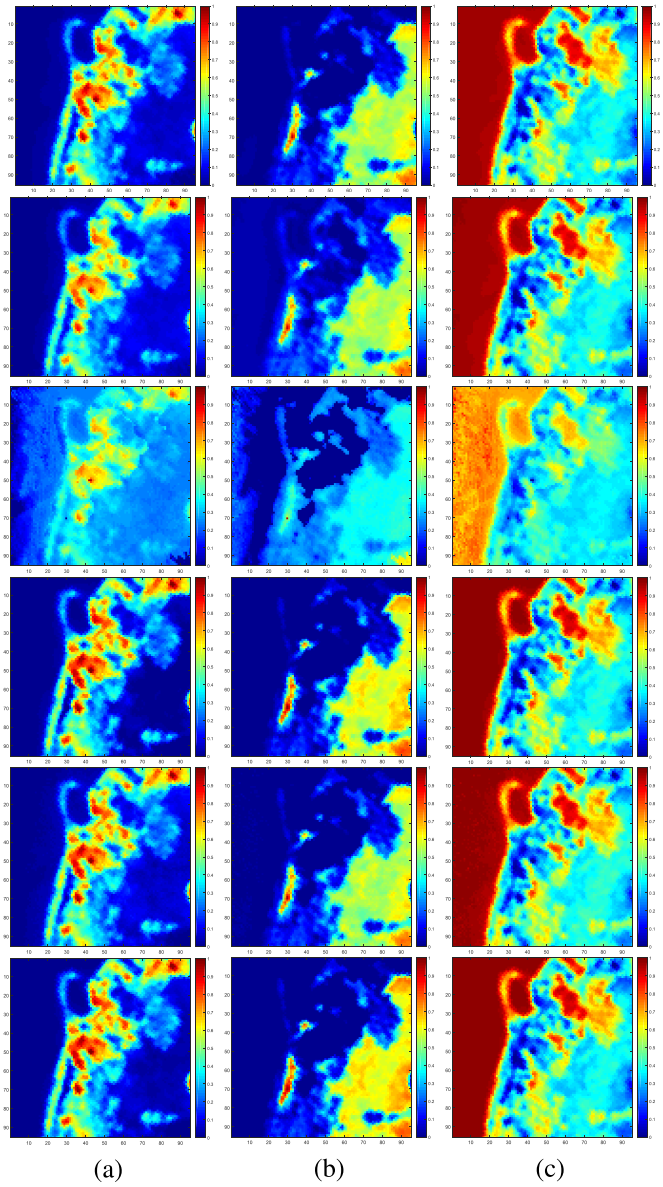


(a) (b) (c)

Fig. 5. Real hyperspectral images, i.e. Samson, Jasper Ridge and AVIRIS Cuprite. (a) Samson. (b) Jasper Ridge. (c) AVIRIS Cuprite.

$952 \times 952$ pixels and each pixel has 156 spectral bands in the range of [0.41 0.889] $\mu$m, but researchers only use a cropped region to reduce computational cost in most cases. This region starts from the pixel located at coordinate (252, 332), and has the size of $95 \times 95$. According to previous researches, this region contains three materials, i.e., tree, water, and soil. Table VIII lists the compared results in terms of estimated endmembers. We can see that the proposed method achieves lower SAD values for all endmembers, which denote the superiority of ARP-NMF. Fig. 6 shows the visual comparison of abundance maps estimated by all algorithms. Due to the ground-truth abundance maps are unavailable, we compare the estimated abundance maps with the real scene image, i.e. Fig. 5(a). From Fig. 5(a), we can see that the bottom of the right side is covered by soil. And in abundance maps, the higher value means the proportion of the material is larger. It can be seen in Fig. 6(b) that the color-related soil estimated by ARP-NMF is closer to red, which means the abundance map of soil estimated by ARP-NMF is closer to the actual scene. In addition, it can be seen that the differences between estimated abundances maps by using $L_{1/2}$-NMF, RCo-NMF, GL-NMF and $HGL_{1/2}$-NMF are very small. Therefore, we list the average $RMSE_X$ values to present the quantitative comparison in Table IX. Combining the experimental results shown in Tables VIII and IX, we can see that the ARP-NMF method achieves the best unmixing performance.

*2) Jasper Ridge Data:* In the hyperspectral unmixing field, the Jasper Ridge data set is also widely used by researchers. It has $512 \times 614$ pixels, and each pixel contains 224 bands covered the range from 380 to 2500 nm. In this experiment, we only consider a subimage sized of $100 \times 100$. The first pixel in the subimage starts from the pixel located at the coordinate of (105, 269) in the original image. In order to eliminate the effects of water vapor and atmosphere, we remove the bands of 1–3, 108–112, 154–166. That is to say, 198 bands remained for each pixel in this experiment. The experimental results are shown in Tables X, XII, and Fig. 7.



(a) (b) (c)

Fig. 6. Abundance maps estimated by $L_{1/2}$-NMF, RCo-NMF, GLr-NMF, GL-NMF, $HGL_{1/2}$-NMF and ARP-NMF method, respectively, from the first row to the last bottom in turn on Samson data set. (a) Tree. (b) Soil. (c) Water.

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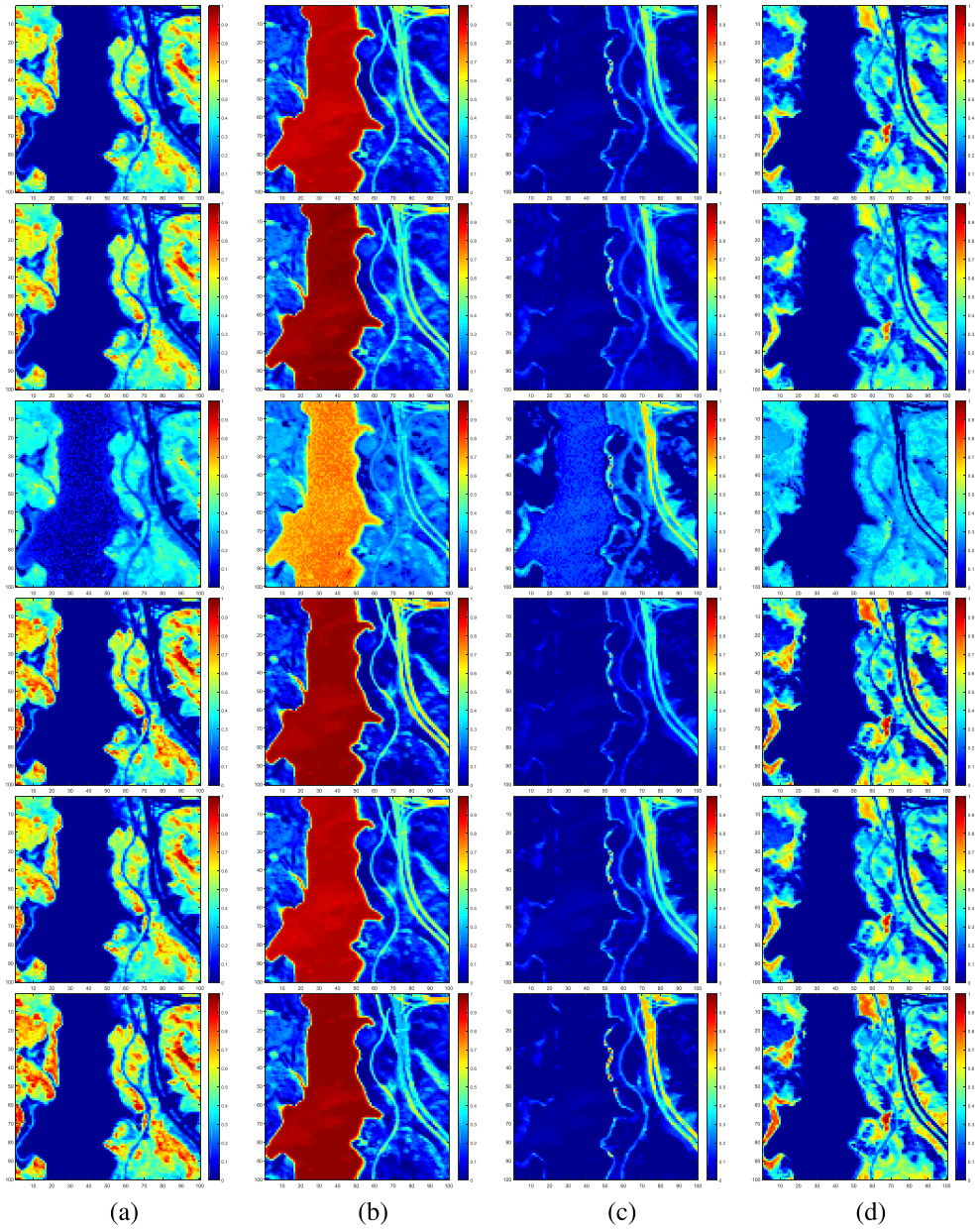



Fig. 7. Abundance maps estimated by $L_{1/2}$-NMF, RCo-NMF, GLr-NMF, GL-NMF, $HGL_{1/2}$-NMF and ARP-NMF, respectively, from the first row to the last in turn on Jasper Ridge data set. (a) Tree. (b) Water. (c) Road. (d) Soil.

TABLE IX
RMSE$_X$ AND STANDARD DEVIATION OF THE SIX METHODS ON THE SAMSON DATA

| Method | RMSE$_X$ |
|---|---|
| $L_{1/2}$-NMF | 0.1455±0.0015 |
| RCo-NMF | 0.1351±0.0018 |
| GLr-NMF | 0.8025±0.0086 |
| GL-NMF | 0.1642±0.0321 |
| $HGL_{1/2}$-NMF | 0.1489±0.0071 |
| ARP-NMF | **0.1180±0.0013** |

Fig. 7 displays a visual comparison among the abundance maps estimated by six methods. It can be seen from Fig. 5(b) that the white part on the right side is the road. And the color of the road in the estimated abundance map by ARP-NMF is closer to red, which means the abundance map of the road estimated by ARP-NMF is closer to the actual scene. Tables X and XII show the quantitative comparison in terms of average SAD and RMSE$_X$. Form Table X, we can see that the proposed ARP-NMF method obtain the lowest SAD values for all endmembers. In particular, the mean SAD value of ARP-NMF is lower about 0.4 than that of $L_{1/2}$-NMF, RCo-NMF and $HGL_{1/2}$-NMF methods. In addition, in the view of data reconstruction, the ARP-NMF method achieves the lowest average RMSE$_X$ value than other methods, which also confirmed its great unmixing performance.

Note that the experiments of six algorithms on the Jasper Ridge data are implemented by using MATLAB R2016 on a Windows system with a 3.40-GHz Intel Core i7 CPU and 16 G RAM. The running time of different algorithms is

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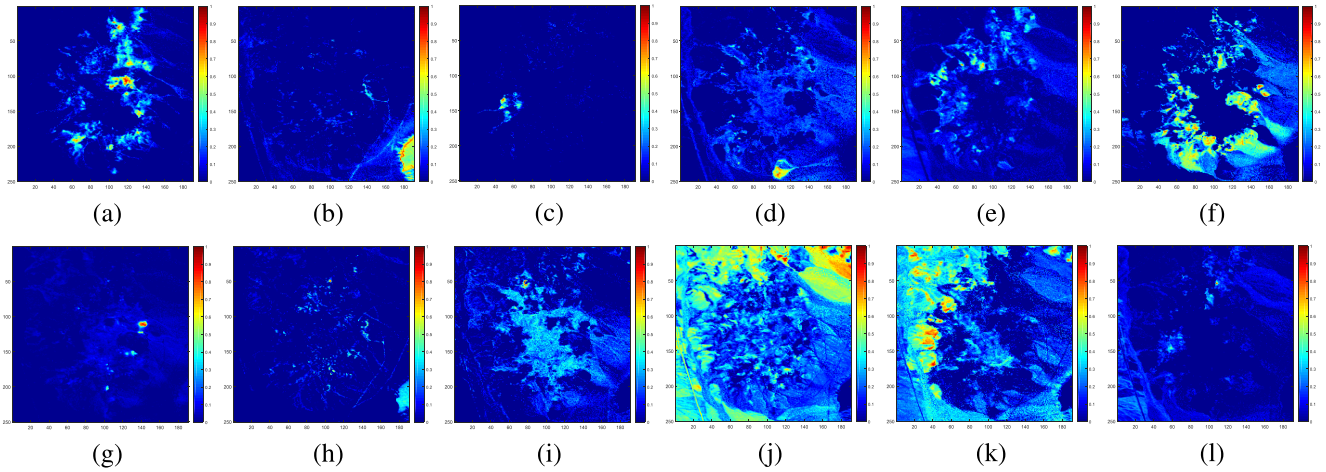



Fig. 8. Abundance maps of 12 endmembers obtained by ARP-NMF method on the AVIRIS Cuprite data set. (a)Alunite. (b)Andradite. (c) Buddingtonite. (d) Dumortierite. (e) Kaolinite1. (f) Kaolinite2. (g) Muscovite. (h) Montomorillonite. (i) Nontronite. (j) Pyrope. (k) Sphene. (l) Chalcedony.

TABLE X
SAD AND STANDARD DEVIATION OF THE SIX METHODS ON THE JASPER RIDGE DATA

| Endmember | $L_{1/2}$-NMF | RCo-NMF | GLr-NMF | GL-NMF | $HGL_{1/2}$-NMF | ARP-NMF |
|---|---|---|---|---|---|---|
| Tree | 0.1738±0.0125 | 0.1728±0.01115 | 0.2364±0.0550 | 0.1432±0.0780 | 0.1559±0.0309 | **0.1091±0.0334** |
| Soil | 0.3541±0.0232 | 0.3084±0.0111 | 0.3567±0.0945 | 0.3546±0.0066 | 0.3552±0.0121 | **0.2976±0.0337** |
| Water | 0.1422±0.0471 | 0.1816±0.0129 | 0.1482±0.0487 | 0.1578±0.0179 | 0.1395±0.0255 | **0.1031±0.0252** |
| Road | 0.0867±0.0845 | 0.0907±0.0136 | 0.1092±0.1200 | 0.1157±0.0061 | 0.1022±0.0891 | **0.0706±0.0131** |
| Mean | 0.1892±0.1282 | 0.1867±0.1596 | 0.2126±0.1368 | 0.1928±0.1101 | 0.1882±0.1200 | **0.1451±0.0938** |

TABLE XI
RUNNING TIME (IN SECOND) OF THE SIX METHODS ON JASPER RIDGE DATA

| | $L_{1/2}$-NMF | RCo-NMF | GLr-NMF | GL-NMF | $HGL_{1/2}$-NMF | ARP-NMF |
|---|---|---|---|---|---|---|
| Run time | 74.34 | 39.74 | 215.1 | 589.3 | 777.4 | $2.975\times10^3$ |

TABLE XII
$RMSE_X$ AND STANDARD DEVIATION OF THE SIX METHODS ON THE JASPER RIDGE DATA

| Method | $RMSE_X$ |
|---|---|
| $L_{1/2}$-NMF | 0.2106±0.0460 |
| RCo-NMF | 0.2716±0.0124 |
| GLr-NMF | 0.7972±0.1380 |
| GL-NMF | 0.2050±0.0198 |
| $HGL_{1/2}$-NMF | 0.2327±0.0398 |
| ARP-NMF | **0.1866±0.0070** |

displayed in Table XI. From Table XI, it is obvious that the proposed ARP-NMF method needs more running time than other methods. Therefore, in future work, we will devote ourselves to developing a more effective algorithm for hyperspectral unmixing, aiming at obtaining real-time results and better performance.

*3) AVIRIS Cuprite Data:* The third experiment of real hyperspectral data is on the AVIRIS Cuprite data, which is a benchmark data set used to investigate the performance of the unmixing algorithm on large and complicated data set. This real image contains 250 × 191 pixels and 14 types of minerals. Each pixel includes 224 bands that covered a wavelength range of [0.37–2.48] $\mu$m. In this experiment, we remove the bands severely affected by noise (1–2 and 221–224) and water absorption (104–113 and 148–167) and remain 188 bands to perform unmixing. In addition, as most experiments on this real data assume, we reduce the number of endmembers to 12 by ignoring minor differences between similar minerals. Table XIII shows the SAD values between estimated endmembers and their ground-truth spectral signatures from the USGS library. It can be seen that the endmembers estimated by the ARP-NMF method have better consistency with their corresponding ground-truth for most minerals. Fig. 8 displays the abundances maps of 12 endmembers obtained by the ARP-NMF method. In addition, the comparison of different methods in terms of $RMSE_X$ are listed in Table XIV.

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TABLE XIII
SAD AND STANDARD DEVIATION OF THE SIX METHODS ON THE AVIRIS CUPRITE DATA

| Endmember | $L_{1/2}$-NMF | RCo-NMF | GLr-NMF | GL-NMF | $HGL_{1/2}$-NMF | ARP-NMF |
|---|---|---|---|---|---|---|
| Alunite | 0.1218±0.0121 | **0.1117±0.0076** | 0.1280±0.0116 | 0.1190±0.0108 | 0.1199±0.0090 | 0.1120±0.0083 |
| Andradite | 0.0965±0.0137 | 0.1135±0.0081 | 0.1032±0.0114 | 0.0954±0.0094 | 0.0931±0.0101 | **0.0730±0.0103** |
| Buddingtonite | 0.0898±0.0188 | 0.1281±0.0161 | 0.0954±0.0137 | **0.0767±0.0080** | 0.0859±0.0139 | 0.0775±0.0220 |
| Dumortierite | 0.1076±0.0247 | 0.1283±0.0141 | 0.1155±0.0156 | 0.1067±0.0150 | 0.1052±0.0141 | **0.0792±0.0115** |
| Kaolinite1 | 0.1067±0.0180 | 0.1128±0.0175 | 0.1194±0.0192 | 0.1034±0.0156 | 0.1015±0.0154 | **0.0916±0.0152** |
| Kaolinite2 | 0.0944±0.0195 | 0.1110±0.0165 | 0.1076±0.0207 | 0.0877±0.0219 | 0.0833±0.0132 | **0.0651±0.0152** |
| Muscovite | 0.1128±0.0258 | 0.1398±0.0229 | 0.1262±0.0215 | 0.1164±0.0354 | 0.1087±0.0179 | **0.1021±0.0212** |
| Montmorillonite | 0.1077±0.0308 | 0.1380±0.0414 | 0.1076±0.0265 | 0.0978±0.0333 | **0.0907±0.0218** | 0.0911±0.0233 |
| Nontronite | 0.1301±0.0291 | 0.1594±0.0330 | 0.1338±0.0301 | 0.1310±0.0331 | 0.1156±0.0243 | **0.1043±0.0273** |
| Pyrope | 0.0992±0.0276 | 0.1646±0.1098 | 0.1235±0.0607 | 0.1120±0.0688 | 0.1014±0.0239 | **0.0831±0.0326** |
| Sphene | 0.2606±0.0998 | 0.3153±0.0777 | 0.2708±0.0703 | 0.2932±0.0564 | 0.2721±0.0358 | 0.2084±0.0826 |
| Chalcedony | 0.1911±0.0380 | 0.2154±0.0312 | 0.1828±0.0435 | 0.2162±0.0299 | 0.1916±0.0390 | **0.1590±0.0352** |
| Mean | 0.1265±0.0581 | 0.1532±0.0709 | 0.1345±0.0631 | 0.1296±0.0680 | 0.1224±0.0569 | **0.1030±0.0501** |

TABLE XIV
$RMSE_X$ AND STANDARD DEVIATION OF THE SIX METHODS ON THE AVIRIS CUPRITE DATA

| Method | $RMSE_X$ |
|---|---|
| $L_{1/2}$-NMF | 0.1262±0.0234 |
| RCo-NMF | 0.1432±0.0425 |
| GLr-NMF | 0.3275±0.0454 |
| GL-NMF | 0.1004±0.0374 |
| $HGL_{1/2}$-NMF | 0.1032±0.0284 |
| ARP-NMF | **0.0804±0.0189** |

Overall, the proposed method achieves the best unmixing performance in AVIRIS Cuprite data set.

## V. CONCLUSION

In this article, an adaptive relationship preserving-based sparse NMF method is proposed. This method regards similarity learning of pixels and unmixing as an alternative optimization process. Through this process, the proposed method can obtain the optimal similarity weights for unmixing. In addition, our ARP-NMF method utilizes spectral and spatial local structure to make each pair of abundances share a consistent similarity degree with their corresponding pixels. In other words, the proposed method effectively keeps the structure consistency between pixels and abundances, which contributes to improving the performance of unmixing. Experimental results both on the synthetic data and the real hyperspectral data sets reveal that our proposed method has better generalization ability for diverse hyperspectral images than several representative unmixing methods.

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